MODEL STRATEGY & HYPERPARAMETERS

Physical-Layer Anomaly Detection in LoRaWAN Networks

Comprehensive breakdown of classifier strategies, exact model parameters, and leak-free evaluation results on physical RF metrics.

RF Feature Engineering

Zero Data Leakage Protocol

ARCHITECTURAL TAXONOMY

Model Strategy & Learning Paradigms

Random Forest

Supervised Baseline

Primary ensemble baseline for RF feature classification.

Focus: Feature variance capture

XGBoost

Supervised High-Capacity

Gradient boosting tuned for imbalanced rogue packet detection.

Focus: High precision detection

Isolation Forest

Unsupervised Anomaly

Trained exclusively on normal traffic for zero-day rogue detection.

Focus: Novelty isolation

CONFIGURATION & PARAMETERS

Model Hyperparameter Specifications

Model	Key Hyperparameters	Value / Setting	Engineering Context
Random Forest	n_estimators / max_depth	**100 / 10**	Prevents overfitting on device features
	min_samples_split / criterion	**5 / 'gini'**	Optimal node splitting variance threshold
XGBoost	learning_rate / max_depth	**0.1 / 6**	Controlled boosting step size
	scale_pos_weight / subsample	**9.8 / 0.8**	Balanced for ~9.1% rogue packet imbalance
Isolation Forest	contamination rate	**0.09**	Aligned with baseline network anomaly ratio
	max_samples / n_estimators	**'auto' (256) / 100**	Sub-sampling for fast anomaly isolation

HELD-OUT BENCHMARKS

Model Performance & Evaluation Metrics

PRIMARY CLASSIFIER F1-SCORE

98.8%

XGBoost peak balanced accuracy

FALSE POSITIVE RATE (FPR)

0.1%

Prevents dropping clean node traffic

ROC-AUC SEPARATION

0.999

Near-perfect class separability

Metric	XGBoost (Primary)	Random Forest (Baseline)	Isolation Forest (Unsupervised)
Precision	**99.1%**	98.4%	88.5%
Recall	**98.6%**	97.2%	91.2%
F1-Score	**98.8%**	97.8%	89.8%
False Positive Rate (FPR)	**0.1%**	0.3%	2.1%
ROC-AUC	**0.999**	0.995	0.954